

Project title: CareBook — Patient Appointment Booking System

Date: June 2025

Prepared by: Aakash Chourasiya

Version: 1.0

1. Problem statement:

RdxCare Private Clinic, Indore, currently manages all patient appointments through phone calls and a manual register maintained by receptionists.

- This process leads to frequent double bookings, missed follow-up appointments, and long wait times for patients.
- Doctors have no visibility into their daily schedule until they arrive at the clinic.
- There is no digital record of appointment history, making it difficult to track patient visits over time.

The clinic requires a web-based patient appointment booking system that allows patients to self-book appointments online, gives doctors a real-time schedule view, and enables receptionists to manage and override bookings when needed.

2. Project scope:

IN SCOPE

Patient registration and login
Online appointment booking
Doctor schedule management
Appointment cancellation and rescheduling
Email and SMS confirmation notifications
Receptionist admin panel

OUT OF SCOPE

Payment gateway / billing
Mobile app (web only for v1)
Electronic Health Records (EHR)
Multi-clinic / multi-branch support
Video consultation / telemedicine
Insurance claim management

3. Stakeholder register:

STAKEHOLDER	ROLE	GOAL	PAIN POINT
Patient	End user	Book appointments without calling or waiting	Long hold times, no visibility of available slots
Doctor	Internal user	See daily schedule in advance, reduce no-shows	No schedule visibility until arriving at clinic
Receptionist	Admin user	Manage all bookings efficiently, reduce phone load	Manual register is error-prone, no backup
Clinic Admin	Business owner	Reduce missed appointments, improve patient satisfaction	No data on appointment trends or doctor utilization

3. Requirement gathering approach:

STAKEHOLDER INTERVIEW QUESTIONS -

Patient questions

1. How do you currently book appointments?
2. What would make the process easier?
3. Do you prefer email or SMS reminders?

Doctor questions

1. How far in advance do you want to see your schedule?
2. Should patients be able to book with any available doctor or their regular one only?

Receptionist questions

1. How many appointments does the clinic handle daily?
2. Do you need the ability to override a patient's booking?
3. What happens when a doctor is absent?

4. FUNCTIONAL REQUIREMENTS (fr)

ID	REQUIREMENT	PRIORITY
FR-01	System shall allow patients to register using name, mobile number, and email	High
FR-02	System shall allow patients to log in using mobile OTP or email-password	High
FR-03	System shall display available appointment slots by doctor, date, and time	High
FR-04	System shall allow patients to book, reschedule, or cancel an appointment	High
FR-05	System shall send confirmation via email and SMS after successful booking	High
FR-06	System shall send a reminder notification 24 hours before the appointment	Medium
FR-07	System shall provide doctors with a daily schedule view showing all booked appointments	High
FR-08	System shall allow receptionists to create, modify, and cancel appointments on behalf of patients	High
FR-09	System shall prevent double-booking of the same doctor at the same time slot	High
FR-10	System shall maintain a history of all past appointments per patient	Medium
FR-11	Admin shall be able to configure doctor availability (working days, hours, holidays)	Medium
FR-12	System shall generate a daily appointment report for the clinic admin	Low

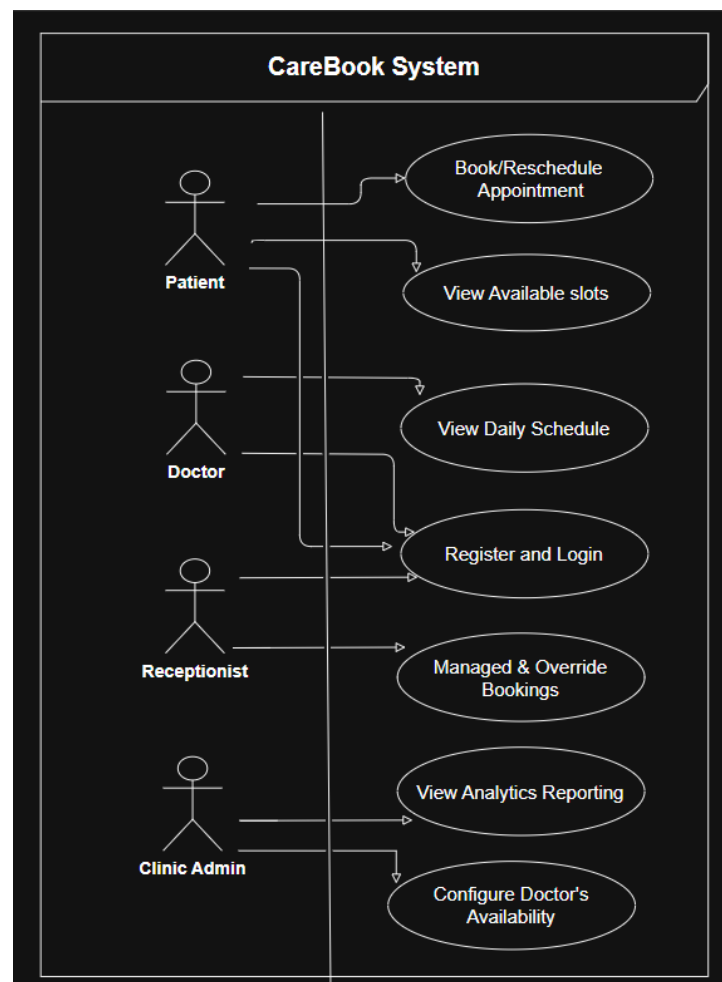
5. NON – FUNCTIONAL REQUIREMENTS (nfr)

ID	REQUIREMENT	CATEGORY	TARGET
NFR-01	System shall load any page within 3 seconds on a standard internet connection	Performance	< 3 sec
NFR-02	System shall be available 99.5% of the time excluding scheduled maintenance	Availability	99.5% uptime
NFR-03	Patient personal data shall be encrypted in storage and during transmission	Security	AES-256 / HTTPS
NFR-04	System shall support up to 500 concurrent users without degradation	Scalability	500 users
NFR-05	System shall be accessible on Chrome, Firefox, and Safari browsers	Compatibility	3 browsers
NFR-06	System UI shall be responsive and usable on mobile screen sizes (min 360px)	Usability	Mobile-friendly

6. Use Case Diagram & Descriptions

Core Use Case Description: Book an Appointment

FIELD	DESCRIPTIONS
Use Case ID	UD-03
Use Case Name	Book an appointment
Pre-conditions	Patients must be successfully registered and logged into the portal.
Post-conditions	1. Appointment status saved as confirmed in the databases. 2. Selected time slot is blocked for other users. 3. Conformation notifications dispatched.
Main Success Scenario	1. Patients navigate to the booking dashboard. 2. Patients select a doctor and a preferred date. 3. System. renders available 20-minute intervals. 4. Patients input the reasons for the visit and click confirms. 5. System confirms availability, reserve the slot and display a success screen.
Extension	3a. Slots simultaneously booked by another user. System triggers a real-time validation error- "Slot no longer available, please select another time."



7. Interface Wireframes & Element Mapping

Wireframe Mockup (The Visual)

Element & Functional Mapping Table

SCREEN AREA	UI ELEMENT	ASSOCIATED REQUIREMENTS	FUNCTIONAL RULE/ BUSINESS LOGIC
Header	Navigation Links & Avatar	FR-01, FR-02, FR-04	Provides a role-based navigation and minimal logout capability without disrupting intrusive modals.
Step 1 drop down.	Searchable Select Box	FR-03, FR-09	Fetches and present active doctors filtered dynamically by specializations.
Step 3 Grid	20-min Interactive Buttons	FR-03, FR-09	State Available: Background flat gray with pointer-events deactivated to prevent double-booking concurrency errors (FR-09).
Step 4 Input	Text Field (Max 250 chars)	FR-04	Allows free-form notation of symptoms with input validation strings.
Footer	Security Compliance text	NFR-03	Display structural operational build metadata (v1.0.4) and trust seals indicating active AES-256 layer encryption.

8. USER STORIES WITH ACCEPTANCE CRITERIA

US1: Patient Registration

As a new patient, I want to register on the platform using my mobile number and email, so that I can create an account and start booking appointments.

Acceptance criteria:

Scenario 1: Successful account creation with OTP verification

- Given a new user is on the registration page,
- When they enter their name, mobile number, email and password and submit the form,
- Then the system sends a 6-digit OTP to their mobile number,
- And when the user enters the correct OTP on the verification screen,
- Then their account is successfully created, and they are redirected to the dashboard homepage.

Scenario 2: Duplicate mobile number rejected at registration

- Given a mobile number already exists in the system,
- When a new user tries to register with the same number,
- Then the system shows an error message saying the number is already registered.

Scenario 3: Invalid OTP blocks account creation

- Given a new user has submitted valid registration details and is on the OTP verification screen,
- When they enter an incorrect or expired OTP and click confirm,
- Then the system rejects the verification request.
- And displays a validation error stating: "Invalid OTP. Please try again or request a new code."

US2: View available slots

As I patient I want to view available slots by doctors and date.
So that I can choose time that suits me.

Acceptance criteria:

Scenario 1: Available slots displayed by doctor and date

- Given a logged-in patient is on the booking page,
- When they select a doctor and a future date,
- Then the system displays all available and unavailable time slots for that doctor.

Scenario 2: Booked slots appear greyed out

- Given a patient is viewing slots,
- When a slot is already booked by another patient,
- Then that slot appears greyed out and cannot be selected.

Scenario 3: Slots update in real-time without page refresh

- Given a patient is viewing slots,
- When another patient books a slot simultaneously,
- Then the slot updates in real-time without a page refresh.

US3: Book an appointment

As a patient, I want to book an appointment with a doctor, so that I can secure my visit without calling the clinic.

Acceptance criteria:**Scenario 1: Successful booking confirmation displayed**

- Given a logged-in patient has selected a doctor, date, time slot and reason for visit,
- When they click Confirm Booking,
- Then the system saves the booking and shows a confirmation screen with all details.

Scenario 2: Confirmation email and SMS sent within 2 minutes

- Given the booking is confirmed,
- When the system processes it,
- Then a confirmation email and SMS are sent to the patient within 2 minutes.

Scenario 3: Slot blocked immediately after booking

- Given a patient confirms a slot,
- When the booking is saved,
- Then that slot is immediately blocked and unavailable to all other patients.

US4: Cancel or reschedule

As a patient, I want to cancel or reschedule an appointment, so that I can manage my visit according to my availability.

Acceptance criteria:**Scenario 1: Patient successfully cancels more than 2 hours before appointment**

- Given a patient has a booked appointment that is more than 2 hours away,
- When they choose to cancel it,
- Then the system cancels the booking and sends a confirmation via email and SMS.

Scenario 2: Cancellation blocked within 2-hour window

- Given a patient tries to cancel an appointment within 2 hours of the scheduled time,
- When they attempt to cancel,
- Then the system shows a message saying cancellation is not allowed within 2 hours.

Scenario 3: Reschedule opens with existing booking pre-filled

- Given a patient selects reschedule,
- When the slot selection screen opens,
- Then their current booking details are pre-filled and they can select a new date and time.

US5: Doctors views daily schedules

As a doctor, I want to view my appointment schedule for the day, so that I can prepare for each patient visit in advance.

Acceptance criteria:**Scenario 1: Doctor sees full appointment list for selected date**

- Given a logged-in doctor is on their dashboard,
- When they open their schedule for a selected date,
- Then they see a list of all appointments showing patient name, time slot and reason for visit.

Scenario 2: Schedule updates automatically on new booking or cancellation

- Given a doctor is viewing their schedule,
- When a patient books or cancels an appointment,
- Then the schedule updates automatically without any manual refresh.

Scenario 3: Doctor marks appointment as Completed

- Given a doctor has completed a visit,
- When they mark the slot as Completed,
- Then the appointment status is updated in the system immediately.

US6: Receptionist Manages Booking

As a receptionist, I want to create and manage appointments on behalf of patients, so that I can assist walk-in or phone patients efficiently.

Acceptance criteria:**Scenario 1: Receptionist finds patient by name or mobile number**

- Given a receptionist is logged into the admin panel,
- When they search for a patient by name or mobile number,
- Then the system displays the matching patient profile and their existing appointments.

Scenario 2: All receptionist actions logged with timestamp

- Given a receptionist creates, modifies or cancels an appointment,
- When the action is saved,
- Then the change is logged with the receptionist's username and a timestamp.

Scenario 3: Patient notified of any receptionist-initiated change

- Given a receptionist makes a change to any appointment,
- When the system processes it,
- Then the patient receives an SMS and email notification about the change.

US7: Appointment reminder notification

As a patient, I want to receive a reminder before my appointment, so that I do not miss my scheduled visit.

Acceptance criteria:**Scenario 1: Reminder sent 24 hours before appointment**

- Given a patient has a confirmed appointment,
- When 24 hours remain before the appointment time,
- Then the system automatically sends a reminder via SMS and email with doctor name, date, time and clinic address.

Scenario 2: Patient cancels directly from reminder message

- Given the reminder is sent
- When the patient clicks the cancel link in the reminder message
- Then the appointment is cancelled and the slot is released.

Scenario 3: Reminder delivery status logged as delivered or failed

- Given a reminder is triggered,
- When the system attempts delivery,
- Then the delivery status is logged as either delivered or failed

US8: Admin configures doctor availability

As a clinic admin, I want to configure each doctor's working hours and days off, so that patients can only book during available slots.

Acceptance criteria:**Scenario 1: Doctor availability reflects immediately in booking calendar**

- Given a clinic admin is on the doctor management page,
- When they set working days, hours and any blocked dates for a doctor,
- Then those settings are immediately reflected in the patient-facing booking calendar.

Scenario 2: Admin alerted when blocking a date with existing bookings

- Given an admin blocks a date for a doctor that already has existing bookings,
- When the block is saved,
- Then the system triggers an alert to the admin listing all affected appointments.

US9: Doctor search and filter

As a patient, I want to search doctor by specialization, name and availability, so that I can quickly find the right doctor for my medical need and book an appointment efficiently.

Acceptance criteria:

Scenario 1: Patient searches doctor by name in real time

- Given a patient is on the 'Find a Doctor' page,
- When they type a name in the search bar,
- Then the system displays all doctors whose names match the search text in real time.

Scenario 2: Doctors filtered by specialization dropdown

- Given a patient selects a specialization from the dropdown,
- When the filter is applied,
- Then only doctors with that specialization are shown in the results.

Scenario 3: No results message shown when filters find no match

- Given no doctors match the selected filters.
- When the results area loads,
- Then the system shows a "No doctors found" message with an option to clear all filters.

US10: Admin Reporting Dashboard

As a clinic admin, I want to access a reporting dashboard showing appointment trends and doctor utilization, so that I can make data-driven decisions to improve clinic operations and patient satisfaction.

Acceptance criteria:

Scenario 1: Dashboard displays appointment metrics for selected date range

- Given a clinic admin is on the Reports page,
- When they select a date range (daily, weekly, monthly or custom),
- Then the dashboard displays total appointments, completed, cancelled and no-show counts for that period.

Scenario 2: Doctor utilization shown per selected period

- Given the admin wants to assess doctor performance,
- When they view the doctor utilization section,
- Then they see each doctor's appointment count for the selected period.

Scenario 3: Peak hours and busiest days highlighted

- Given the admin wants to plan staffing,
- When they view the peak hours section,
- Then the dashboard highlights the busiest time slots and days of the week based on historical data.